

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AI & ML / 2024	Date:	

Code of Conduct

I V Yogeswar / 192425175 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V Yogeswar

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

He → Ravi / Arun

him → Ravi / Arun

it → book

they → Ravi and Arun

the book → the book mentioned earlier

b) Apply the following constraints to resolve the references:

- **Gender and number agreement** :- Ravi and Arun are both singular and male, so both can grammatically match He and him.
- **Recency** :- Arun is more recent than Ravi, so recency alone favors Arun for He.
- **Grammatical role** :- Ravi is the subject of the previous sentence and is the continuing discourse topic, which favors Ravi as the antecedent of He.
- **Semantic compatibility** :- Ravi can look for a book, Arun can help Ravi find it, and both can discuss the book.
- **Discourse coherence** :- Interpreting He as Ravi creates a coherent sequence in which Ravi is looking for a book and Arun helps him find it.

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	Valid	Gender and number agree; Ravi is the continuing discourse subject/topic.
He	Arun	Less preferred	Gender and number agree, but grammatical-role continuity favors Ravi.
him	Ravi	Valid	Arun can help Ravi find the book; semantically coherent.
him	Arun	Invalid/less preferred	Arun helping himself is less coherent in this context.
it	book	Valid	it refers to the book being found.
it	library	Invalid	A library is not the object being found.
they	Ravi and Arun	Valid	Plural pronoun refers to both people.

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
the book	book	Valid	Refers back to the previously mentioned book.

d) Provide the final coreference chains.

Ravi → He → him → they

Arun → they

book → it → the book

e) Rewrite the paragraph by replacing pronouns with their correct references.

"Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving."

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

If the sentence appeared before Arun was introduced, there would be no clearly established antecedent for He within the discourse. The system would have to use previous discourse context or external information to determine who He refers to, making reference resolution more uncertain.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.
5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

a) Required Dialog Acts

Reassure + Advise

b) Important Entities

- assignment
- mistakes
- you

These entities should remain consistent throughout the response.

c) Three Possible Responses

Response 1:

"Don't worry; you can review the assignment and identify any mistakes. Use the feedback to improve your work and feel more confident."

Response 2:

"It's normal to worry about mistakes after submitting an assignment, so try to stay confident. You can review the assignment later and use the feedback to improve your future work."

Response 3:

"You have already submitted the assignment, so don't be too worried about possible mistakes. Review your work when possible, and use the feedback to improve your understanding and become more confident."

d) Comparison

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
Response 1	High	High	High	High
Response 2	High	High	High	High
Response 3	High	High	High	High

e) Best Response

Response 1 is the best response because it directly reassures the student and provides practical advice. It maintains entity coherence, uses the required keywords review, mistakes, feedback, improve, and confident, and maintains a positive and polite tone.

f) Effect of Losing Entity Coherence

If entity coherence is not maintained, the response may refer to unrelated objects or actions. This can make the response confusing, irrelevant, and difficult for the user to understand.

3. Word Sense Disambiguation and Meaning Representation

a) Possible Meanings of "Bank"

Sense 1: A financial institution.

Sense 2: The side of a river.

b) Correct Sense

The correct sense is:

Bank = the side of a river.

The words boats, water, and moving across the water strongly indicate a river or water-related location. Therefore, the financial-institution sense is incompatible with the surrounding context.

c) Predicate Logic Representation

Let:

- **Priya(x) = x is Priya**
- **RiverBank(y) = y is a river bank**
- **Boat(z) = z is a boat**
- **Water(w) = w is water**
- **Sit(x) = x sits**
- **Beside(x,y) = x is beside y**
- **Watch(x,z) = x watches z**
- **Move(z) = z moves**

Predicate representation:

$\exists x \exists y \exists z \exists w$ [

Priya(x) $\wedge$ RiverBank(y) $\wedge$ Boat(z) $\wedge$ Water(w) $\wedge$

Sit(x) $\wedge$ Beside(x,y) $\wedge$ Watch(x,z) $\wedge$ Move(z) $\wedge$

Across(z,w)

]

This represents that Priya is sitting beside the river bank and watching boats that are moving across the water.

d) Unambiguous Paraphrase

"Priya sat beside the river and watched the boats moving across the water."

e) WSD and Machine Translation

Word Sense Disambiguation improves Machine Translation by determining the intended meaning of an ambiguous word before translation. This helps the translation system select the correct target-language word and prevents contextually incorrect translations. For example, correctly interpreting bank as a river bank instead of a financial institution ensures that the translated sentence preserves the intended meaning.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	3	3	3	3		16	73
2	3	3	3	3		16	
3	3	3	3	3		16	

Faculty In-charge	Course Coordinator	Program Director
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Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____